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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

CS3101-1- COMPILER DESIGN
(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is the primary purpose of a symbol table in a compiler?**
A) To store intermediate code during code generation
B) To maintain information about identifiers and their attributes
C) To detect and report syntax errors
D) To optimize the target code during compilation
2. **Which component of a simple compiler is responsible for reading the source code and grouping characters into tokens?**
A) Syntax Analyzer
B) Lexical analyzer
C) Semantic analyzer
D) Code generator
3. **Which phase of the compiler involves generating machine-independent code?**
A) Target code generation
B) Code optimization
C) Intermediate code generation
D) Syntax analysis
4. **What is the role of regular expressions in lexical analysis?**
A) Generate parse trees
B) Define the syntax rules of the language
C) Specify patterns for tokens
D) Optimize the intermediate code
5. **What is the primary purpose of input buffering in lexical analysis?**
A) To store tokens for syntax analysis
B) To reduce the number of I/O operations and minimize backtracking
C) To handle syntax errors in the source code
D) To simplify semantic analysis
6. **In LEX, how do you define a pattern that matches a word starting with 'a' and ending with 'z'?**
A) a.*z
B) a*.z
C) [a-z]*
D) az*
7. **Which of the following best describes a lexeme?**
A) A rule in LEX
B) A syntax rule used in parsers
C) A reserved keyword in a programming language
D) The smallest sequence of characters forming a token
8. **Which of the following is a valid LEX token pattern for matching a floating-point number?**
A) [0-9]*.[0-9]+
B) [0-9]+.[0-9]*
C) [0-9]*\.[0-9]+
D) \d+\.\d*
9. **In LEX program, what does the yytext variable represent?**
A) The regular expression used for matching tokens
B) The entire input stream being processed
C) The current matched token text
D) The output of the parser
10. **The primary role of the parser in a compiler is to:**
A) Check lexical errors
B) Check syntax and build the parse tree
C) Generate intermediate code
D) Optimize the generated code

11. Which of the following parsers is a top-down parser?
 A) SLR parser B) LALR parser
 C) LL parser D) Canonical LR parser
12. What is the primary difference between LL and LR parsers?
 A) LL parsers use top-down parsing, LR parsers use bottom-up parsing B) LL parsers handle left recursion, LR parsers cannot
 C) LR parsers use recursive descent, LL parsers use shift-reduce D) Both are identical
13. Which derivation is generated by the bottom-up parser?
 A) Right-most derivation in reverse B) Left-most derivation in reverse
 C) Right-most derivation D) Left-most derivation
14. If a state does not know whether it will make a shift operation or reduction for a terminal is called ...
 A) Reduce conflict B) Reduce /Reduce conflict
 C) Shift conflict D) Shift/reduce conflict
15. The context free grammar $S \rightarrow SS \mid 0S1 \mid 1S0 \mid \epsilon$ generates
 A) An unequal number of 0's and 1's B) An equal number of 0's and 1's
 C) The set of all binary strings with more 1s than 0s D) The set of all binary strings with more 0s than 1s
16. What is true about Syntax Directed Definitions?
 A) In SDDs, only synthesized attributes are allowed, not inherited attributes. B) A syntax-directed definition (SDD) is a Context Free Grammar together with attributes and rules
 C) SDDs can be used only for top-down parsing techniques. D) SDDs are not used in compiler design.
17. Which parser is most powerful among the following parsers?
 A) Operator Precedence B) LALR
 C) SLR D) CLR
18. In the case of a bottom-up evaluation of a definition that is syntax-directed, the inherited attributes can:
 A) only be evaluated when the definition consists of synthesized attributes B) never be evaluated
 C) only be evaluated when the definition is L-- attributed D) always be evaluated
19. Which of the following is a primary goal of code optimization in compiler design?
 A) To increase the size of the compiled code B) To reduce the compilation time
 C) To add more comments to the code D) To improve the performance of the generated code
20. In compiler design, dead code elimination refers to:
 A) Removing code that is never executed B) Eliminating code that causes runtime errors
 C) Deleting comments and whitespace from the source code D) Removing code that has been deprecated in the language standard

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the working of different phases of a compiler with an example.	6	L*2	1	1
	b) Write the left factoring algorithm and perform the left factoring of the grammar given below: $A \rightarrow ad \mid a \mid ab \mid abc \mid b$	5	L3	2	1

- c) Construct transition diagram to recognize the tokens described using the following regular definitions:
- $digit \rightarrow [0-9]$
 $digits \rightarrow digit^+$
 $number \rightarrow digits (. digits)? (E [+ -]? digits)?$
- 5 L3 1 1
2. a) Consider the following grammar:
- $S \rightarrow (L) \mid a$
 $L \rightarrow L, S \mid S$
- i. Transform the given grammar to be suitable for LL(1) parsing.
 ii. Construct FIRST and FOLLOW sets.
 iii. Construct LL(1) Parse Table.
- 6 L3 2 1
- b) With the help of algorithm explain the concept of input buffering techniques used in the lexical analysis phase.
- 5 L2 1 1
- c) Show that the following grammar is ambiguous:
- $S \rightarrow iEtS \mid iEtSeS \mid a$
 $E \rightarrow b$
- Also write an equivalent unambiguous grammar for the same.
- 5 L2 2 1
3. a) Given the grammar:
- $S \rightarrow aABb$
 $A \rightarrow c \mid \epsilon$
 $B \rightarrow d \mid \epsilon$
- i. Construct FIRST and FOLLOW sets.
 ii. Construct the predictive parsing table.
 iii. Show the moves made by the predictive parser on the input acdb.
- 6 L3 2 1
- b) For the following grammar give the left most and right most derivation for the string $aa+a^*$
- $S \rightarrow SS + \mid SS^* \mid a$
- 5 L3 2 1
- c) Develop a LEX program to count the number of comment lines in the C program, eliminate them and copy that program into another file.
- 5 L3 1 5
- Unit – II**
4. a) Given the grammar $E \rightarrow (E) \mid id$
 Construct the following:
- i. LR(0) automaton
 ii. SLR(1) parsing table
 iii. Moves made on the string (id)
- 6 L3 3 1
- b) Develop a YACC program to evaluate an arithmetic expression involving operators +, -, * and /
- 5 L3 3 5
- c) Write quadruple, triples, and indirect triples for the following expression:
- $(x + y) * (y + z) + (x + y + z)$
- 5 L3 4 1
5. a) For the given grammar, write annotated parse tree for the input $3 * 5$ using top down approach. Write semantic rule for each step.
- $T \rightarrow F T'$
 $T' \rightarrow * F T'$
 $T' \rightarrow \epsilon$
 $F \rightarrow digit$
- 4 L3 4 1

- b) Construct the LALR parsing table for the following grammar:

$S \rightarrow Aa \mid bAc \mid Bc \mid bBa$

$A \rightarrow d$

$B \rightarrow d$

- c) Compare SLR, LALR, and LR parsers.

8	L3	3	1
4	L2	3	1

6. a) Construct CLR parsing table for the following grammar:

$S \rightarrow Aa \mid bAc \mid dc \mid bda$

$A \rightarrow d$

6	L3	3	1
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- b) Give the SDD for simple desk calculator and draw dependency graph for the expression $1 * 2 * 3 * (4 + 5) n$

5	L3	4	1
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- c) Describe S-attributed and L-attributed definitions with suitable example.

5	L1	4	1
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Unit – III

7. a) Discuss the issues in the design of Code Generator.

6	L2	5	1
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- b) Construct the basic block and flow graph for the following 3-address code:

$t1 = x < y$

if $t1$ goto L1

$t2 = x - y$

$z = t2$

goto L2

L1: $t3 = x + y$

$z = t3$

L2:

5	L3	5	1
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- c) Explain Dead Code elimination method with an example.

5	L2	5	1
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8. a) Write the simple Code Generator Algorithm.

6	L2	5	1
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- b) Construct the DAG for the following basic block. Write this basic block after eliminating the common subexpressions considering variable b is not live on exit:

$a = b + c$

$b = a - d$

$c = b + c$

$d = a - d$

5	L3	5	1
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- c) Write the algorithm for determining the liveness and next-use information for each statement in a basic block.

5	L2	5	1
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
